

**Corporate Governance, Female Board Participation, and Bank Profitability: Evidence
from Bangladesh**

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Abstract

This study examines the relationship between corporate governance, women's participation in governance, and the profitability of selected private commercial banks in Bangladesh. The study uses bank-level information for nine major private commercial banks covering 2016–2024.

Profitability is assessed mainly through Return on Assets (ROA), Return on Equity (ROE), and Net Interest Margin (NIM), while women's participation is represented by female board membership. Corporate governance scores and bank size are also considered in the analysis. The study combines descriptive analysis, correlation analysis, and the regression results reported in the submitted dataset.

The descriptive evidence shows substantial differences across banks. Some banks experienced stronger profitability while female representation increased, whereas other banks recorded declining profitability despite relatively stable or improving governance scores. This indicates that female representation alone cannot explain profitability differences across Bangladeshi banks. The reported regression results also require careful interpretation. The coefficient on the female participation variable is negative and statistically insignificant at the conventional 5% level ($p = 0.2069$), while the squared female-participation term is positive and marginally significant at the 10% level ($p = 0.0672$). ROA is strongly associated with the reported dependent measure in the regression, whereas the reported coefficients for NIM, the corporate governance score, and bank size are not statistically significant at the 5% level. Therefore, the findings provide evidence of a possible non-linear relationship, but they do not establish a simple causal claim that increasing the number of women on bank boards automatically increases profitability.

Overall, the study suggests that women's participation is an important governance issue in Bangladesh, but its financial effect should be understood together with bank-specific factors, governance quality, operating efficiency, and the wider economic and regulatory environment. The paper also identifies data-quality issues in the submitted dataset that should be verified against the original annual reports before final submission or publication.

1. Introduction

Corporate governance refers to the system through which an organization is directed, monitored, and held accountable. In the banking sector, governance is especially important because banks manage depositors' money, take financial risks, and play a major role in overall financial stability. Good governance can improve transparency, strengthen monitoring, support responsible decision-making, and reduce conflicts between managers and shareholders (OECD, 2021).

At the same time, the composition of corporate boards has become an important part of the governance discussion. Women have increased their participation in higher education and employment, but their representation in senior decision-making positions remains comparatively limited. Previous studies suggest that board diversity can improve monitoring and introduce different perspectives into strategic decision-making, although the empirical evidence on profitability is not uniform (Hillman et al., 2007; Kılıç & Kuzey, 2016).

The question is particularly relevant for Bangladesh. The banking sector has expanded considerably, but women remain a minority in many boardrooms and senior governance positions. Earlier research on Bangladesh reports that women's representation on bank boards has increased over time, but participation remains below that of men (Karim et al., 2022; TBS News, 2022). The important issue is therefore not only how many women are present, but also whether their participation is associated with better governance and financial performance.

International evidence is mixed. Some studies find positive links between female leadership or board diversity and bank performance, while others find no statistically significant relationship. For example, research on Vietnamese banks suggests that female leaders may improve profitability and stability, while board-level female representation by itself does not

always produce the same result (Hoang et al., 2021). Studies of corporate boards similarly indicate that the effect of gender diversity can depend on context, governance culture, and the degree of actual influence held by board members.

This study therefore focuses on nine private commercial banks in Bangladesh over the period 2016–2024. It examines financial performance, corporate governance, bank size, and female participation in governance. Instead of assuming that women’s participation must have a positive effect, the paper compares the patterns in the data and uses the reported statistical results to assess whether the relationship is positive, negative, mixed, or potentially non-linear.

1.1 Research Gap

Existing studies provide useful evidence on corporate governance and gender diversity, but relatively limited work has focused specifically on the relationship between women’s participation in the governance of Bangladeshi private commercial banks and profitability using a multi-year bank-level dataset. Much of the existing discussion concentrates on representation rather than whether female participation is associated with measurable differences in profitability. This study attempts to contribute to that discussion by combining governance, financial-performance, and board-composition information for a multi-year sample.

1.2 Objectives of the Study

- To examine the relationship between corporate governance and the financial performance of selected private commercial banks in Bangladesh.
- To assess the relationship between women’s participation in corporate governance and bank profitability.
- To identify patterns in female board participation across the selected banks during 2016–2024.

- To discuss institutional and structural issues that may affect women's participation in banking governance.
- To provide practical implications for corporate governance and gender inclusion in the Bangladeshi banking sector.

1.3 Research Questions

- How did profitability, governance scores, and bank size change across the selected banks from 2016 to 2024?
- What relationship can be observed between female board participation and bank profitability?
- Does the reported regression provide evidence of a linear or non-linear relationship?
- What factors should be considered before interpreting female representation as a determinant of profitability?

2. Literature Review

Corporate governance is important in banking because information asymmetry, risk-taking, and depositor protection make weak governance potentially costly. Effective boards are expected to monitor managers, challenge excessive risk-taking, and provide strategic guidance. In banking, the board also has a direct role in oversight and risk management (de Andrés & Vallelado, 2008).

Agency theory provides one explanation for why board diversity may matter. Shareholders delegate decision-making authority to managers, which can create conflicts of interest and agency costs. Stronger monitoring can reduce these problems. A board with diverse backgrounds may challenge management from different perspectives and strengthen oversight (Jensen & Meckling, 1976).

Resource dependence theory provides another explanation. Boards do not only monitor managers; they also connect organizations to knowledge, networks, expertise, and external resources. A more diverse board may provide access to a wider range of experiences and relationships that can support organizational decision-making (Pfeffer & Salancik, 1978).

Stakeholder theory broadens the governance perspective by considering employees, customers, regulators, communities, and other stakeholders alongside shareholders (Freeman, 1984). From this perspective, greater representation can signal a more inclusive governance structure. However, representation should not automatically be treated as proof of better financial performance.

Research on gender diversity and corporate performance is mixed. Some studies report positive associations, while other research finds weak or insignificant relationships. Kılıç and Kuzey (2016), for example, examine board gender diversity and firm performance and show that the relationship can differ across performance measures. Reguera-Alvarado et al. (2017) similarly discuss the role of board gender diversity in financial performance. The mixed findings suggest that the effect of gender diversity may depend on firm characteristics, institutional setting, and how effectively directors participate in decision-making.

Within banking, Hoang et al. (2021) report that female leaders can be associated with higher profitability and stability in the Vietnamese banking sector. Galletta et al. (2021) find that gender diversity can also be linked with sustainability performance in banking. These findings support the argument that diversity can add value, but they do not imply that every increase in female representation will produce the same financial outcome in every country.

Bangladesh-specific research is also relevant. Karim et al. (2022) document the development of women's representation on boards of commercial banks in Bangladesh, while

Uddin (2023) examines the effect of board gender diversity on bank performance. These studies indicate that gender diversity is a meaningful governance issue in Bangladesh, but further evidence is needed on how representation relates to profitability over time.

2.1 Theoretical Framework

Agency Theory

Agency theory argues that managers may not always act in the best interests of shareholders. Board monitoring can reduce agency costs. A diverse board may strengthen monitoring by adding different perspectives and questioning management decisions from multiple angles (Jensen & Meckling, 1976).

Resource Dependence Theory

Resource dependence theory views directors as providers of knowledge, networks, reputation, and external connections. Female directors may add experiences and networks that are different from those already represented on the board (Pfeffer & Salancik, 1978).

Stakeholder Theory

Stakeholder theory focuses on the broader group of people affected by corporate decisions. A diverse board can improve representation and may strengthen legitimacy with employees, regulators, customers, and society (Freeman, 1984).

Social Identity Theory

Social identity theory suggests that group composition can influence group interaction. A board containing members from different social groups may consider a wider range of viewpoints. In this study, the theory is used as a possible explanation for differences in board

discussion and decision-making, not as proof that women inherently perform differently from men.

Critical Mass Perspective

The critical-mass argument suggests that a very small minority may have limited influence in a group, while a larger representation can create more meaningful participation. The present dataset contains several boards with two to five women in different years. However, the available regression results are not sufficient to prove a specific three-person threshold. Therefore, the critical-mass idea is treated as a theoretical possibility rather than an established finding of this study.

3. Methodology

The study uses bank-level information for nine private commercial banks in Bangladesh: Dutch-Bangla Bank Limited, BRAC Bank, Bank Asia Limited, City Bank, Prime Bank Limited, Southeast Bank, United Commercial Bank PLC (UCB), Dhaka Bank, and Eastern Bank PLC. The intended study period is 2016–2024.

The source draft states that information was manually extracted from annual reports, financial reports, corporate governance disclosures, and director profiles available through the banks' official websites. The variables include financial performance, female board participation, corporate governance score, and bank size. The study is described as an unbalanced panel because some observations are missing in the submitted dataset.

3.1 Variables and Measurement

Table 1
Variables and Their Measurement

Variable	Type	Measurement used in submitted study
ROA	Dependent	Net income / total assets (%)
ROE	Dependent	Net income / total equity (%)
NIM	Dependent / control	Net interest income / average earning assets (%)
Board Diversity (DR)	Independent	Female directors / total directors x 100
Management Diversity (MR)	Independent	Female senior executives / relevant senior-management positions
Bank Size	Control	Total assets; the draft also describes a log transformation in the variable table
Corporate Governance Score	Control	Compliance-based governance score reported in annual reports
Year Effects	Control	Year indicators intended to capture common time-specific factors

A methodological inconsistency should be noted. The draft alternates between treating NIM as a dependent variable and as a control variable, and it presents a single regression table without clearly identifying the exact dependent variable used for the reported coefficients. The revised version therefore reports the statistical results exactly as supplied, but does not claim that the regression proves a causal relationship.

3.2 Regression Model Reported in the Study

The submitted paper reports the following general panel regression specification:

$$\text{Profitability}_{it} = \alpha + \beta_1 \text{Board Diversity}_{it} + \beta_2 \text{Management Diversity}_{it} + \beta_3 \text{Bank Size}_{it} + \beta_4 \text{CG Score}_{it} + \beta_5 \text{NIM}_{it} + \gamma_t + \varepsilon_{it}$$

The paper also reports a female-participation term and a squared female-participation term in its regression output, suggesting an attempt to test non-linearity. Because the reported specification and the reported coefficient table are not fully aligned, the regression findings should be treated as evidence from the submitted model rather than as a fully reproducible causal estimate.

4. Descriptive Analysis

The submitted descriptive table reports summary values for the selected observations. The statistics show that the banks differ considerably in size and profitability. ROE ranges from 9.5% to 19.7% in the reported summary, while ROA remains close to 1%. Corporate governance scores range from 66.87% to 81.44% in the same summary table, and the reported female-member count ranges from zero to two in those observations.

Table 2
Descriptive Statistics for Key Variables

Variable	Count	Mean	Std. Dev.	Min	Median	Max
ROE	9	14.8	4.27	9.5	16.1	19.7
NIM	9	4.89	0.48	4.10	4.95	5.78
ROA	9	1.13	0.23	0.7	1.1	1.4
Corporate Governance Score	9	70.68	5.11	66.87	70.22	81.44
Bank Size (million BDT)	9	472,633	93,661	264,797	514,399	593,883
Female Members	9	1.11	1.02	0	1	2

The summary statistics should be interpreted carefully because the count of nine is not equal to the full bank-year sample that would result from nine banks observed over nine years.

This suggests that the descriptive table may have been constructed from a selected cross-section or summary series rather than from all available bank-year observations. The paper should therefore not describe these nine observations as if they were the complete 81 bank-year panel.

4.1 Correlation Analysis

Table 3
Correlation Matrix of Independent Variables

Variable	ROE	NIM	ROA	CG Score	Bank Size
ROE	1	0.88	0.72	0.77	0.65
NIM	0.88	1	0.76	0.42	0.88
ROA	0.72	0.76	1	0.85	0.72
CG Score	0.77	0.42	0.85	1	0.65
Bank Size	0.65	0.88	0.72	0.65	1

The reported correlations are positive and in several cases high. For example, the draft reports a correlation of 0.88 between ROE and bank size, 0.85 between ROA and corporate governance score, and 0.85 between NIM and ROA. These values indicate association within the reported data, but correlation does not establish causation. The relatively high correlations also indicate the need to think carefully about multicollinearity when interpreting a multivariable regression.

4.2 Regression Results

Table 4
Regression Results

Variable	Coefficient	Std. Error	t-value	p-value
Intercept	0.1532	2.0210	0.0758	0.9398
Female cumulative	-0.1026	0.0805	-1.2735	0.2069

Female squared	0.00421	0.00227	1.8582	0.0672
NIM	0.0157	0.0725	0.2167	0.8291
ROA	11.1283	0.5390	20.6472	0.0000
Corporate Governance	0.00836	0.0237	0.3522	0.7257
Bank Size	0.00284	0.00192	1.4758	0.1444

The female cumulative term has a negative coefficient (-0.1026) but is not statistically significant at the 5% level ($p = 0.2069$). The squared female term is positive (0.00421) and has $p = 0.0672$, which is suggestive at the 10% level but not statistically significant at the conventional 5% level. Taken together, the reported coefficients are consistent with a possible curved relationship, but the evidence is not strong enough to state that a specific critical mass has been proven.

The coefficient for ROA is positive and highly statistically significant in the reported model ($p < 0.001$). By contrast, the reported NIM, corporate governance score, and bank-size coefficients are not statistically significant at the 5% level. Therefore, the original statement that governance variables have a consistently positive and meaningful effect is too strong for the reported regression output and has been removed from the revised interpretation.

5. Bank-by-Bank Analysis

Dutch-Bangla Bank

Table 5

Financial and Governance Indicators for Dutch-Bangla Bank (2016–2024)

Year	ROE %	ROA %	NIM %	CG Score %	Assets (mn BDT)
2024	9.5	0.7	5.78	70.22	565,529.00
2023	17.9	1.4	4.95	68.15	593,883.10

2022	14.4	1.1	4.1	69.41	555,473.60
2021	16.1	1.1	4.49	67.22	514,399.80
2020	18.4	1.3	4.11	66.87	472,355.40
2019	17.2	1.2	4.85	68.23	390,362.00
2018	19.7	1.3	5.62	69.46	346,468.80
2017	13.2	0.9	5.62	78.49	311,906.80
2016	10.3	0.7	5.69	81.44	264,797.40

Table 6

Board Gender Composition for Dutch-Bangla Bank (2016–2024)

Year	Board Total	Male	Female	Female %
2016	8	8	0	0.0
2017	8	8	0	0.0
2018	8	8	0	0.0
2019	7	6	1	14.29
2020	7	6	1	14.29
2021	7	6	1	14.29
2022	8	7	1	12.5
2023	8	6	2	25.0
2024	8	6	2	25.0

Across the reported period, Dutch-Bangla Bank shows ROE between 9.50% and 19.70% and ROA between 0.70% and 1.40%. Female board representation ranges from 0.00% to 25.00%. The bank-level pattern is therefore mixed: changes in profitability do not move in a uniform direction with female representation, and the governance score also changes from year to year. The figures support comparison, but they do not by themselves prove that board gender composition caused the change in profitability.

Total assets for Dutch-Bangla Bank moved from approximately 264,797 million BDT in 2016 to approximately 565,529 million BDT in 2024, while the reported corporate governance

score ranged from 66.87% to 81.44% over the same period. As with profitability, these year-to-year movements in asset size and governance score do not by themselves indicate whether, or how strongly, female board representation contributed to the bank's financial trajectory.

BRAC Bank

Table 7

Financial and Governance Indicators for BRAC Bank (2016–2024)

Year	ROE %	ROA %	NIM %	CG Score %	Assets (mn BDT)
2024	17.43	1.46	3.02	72.39	930,133.00
2023	17.43	1.13	5.29	80.47	726,723.00
2022	10.22	1.14	5.35	76.02	563,271.00
2021	11.0	1.31	4.6	77.57	449,084.00
2020	10.63	1.18	3.7	78.5	396,982.00
2019	16.29	1.65	4.64	77.63	414,855.00
2018	19.25	1.87	5.13	74.07	358,005.00
2017	22.14	2.02	5.35	97.91	308,317.00
2016	22.16	1.89	5.5	84.0	268,324.00

Table 8

Board Gender Composition for BRAC Bank (2016–2024)

Year	Board Total	Male	Female	Female %
2016	7	5	2	28.57
2017	8	6	2	25.0
2018	9	7	2	22.22
2019	7	4	3	42.86
2020	N/A	N/A	N/A	N/A
2021	10	8	2	20.0
2022	10	8	2	20.0
2023	11	8	3	27.27
2024	9	6	3	33.33

Across the reported period, BRAC Bank shows ROE between 10.22% and 22.16% and ROA between 1.13% and 2.02%. Female board representation ranges from 20.00% to 42.86%. The bank-level pattern is therefore mixed: changes in profitability do not move in a uniform direction with female representation, and the governance score also changes from year to year. The figures support comparison, but they do not by themselves prove that board gender composition caused the change in profitability.

Total assets for BRAC Bank moved from approximately 268,324 million BDT in 2016 to approximately 930,133 million BDT in 2024, while the reported corporate governance score ranged from 72.39% to 97.91% over the same period. As with profitability, these year-to-year movements in asset size and governance score do not by themselves indicate whether, or how strongly, female board representation contributed to the bank's financial trajectory.

City Bank

Table 9

Financial and Governance Indicators for City Bank (2016–2024)

Year	ROE %	ROA %	NIM %	CG Score %	Assets (mn BDT)
2024	26.1	1.7	5.0	89.04	697,349.00
2023	17.6	1.2	4.5	99.96	555,738.00
2022	14.1	1.0	3.7	89.21	506,847.00
2021	15.8	1.2	4.6	89.82	416,902.00
2020	14.8	1.1	3.8	91.08	382,926.00
2019	9.9	0.7	17.2	74.06	354,689.00
2018	8.2	0.7	17.5	75.44	324,780.00
2017	15.9	1.4	14.6	90.09	275,531.00
2016	21.26	1.71	15.8	88.42	25,940.23

Table 10
Board Gender Composition for City Bank (2016–2024)

Year	Board Total	Male	Female	Female %
2016	17	14	3	17.65
2017	17	14	3	17.65
2018	14	11	3	21.43
2019	11	8	3	27.27
2020	12	8	4	33.33
2021	12	8	4	33.33
2022	11	7	4	36.36
2023	11	8	3	27.27
2024	10	7	3	30.0

Across the reported period, City Bank shows ROE between 8.20% and 26.10% and ROA between 0.70% and 1.71%. Female board representation ranges from 17.65% to 36.36%. The bank-level pattern is therefore mixed: changes in profitability do not move in a uniform direction with female representation, and the governance score also changes from year to year. The figures support comparison, but they do not by themselves prove that board gender composition caused the change in profitability.

Total assets for City Bank moved from approximately 25,940 million BDT in 2016 to approximately 697,349 million BDT in 2024, while the reported corporate governance score ranged from 74.06% to 99.96% over the same period. As with profitability, these year-to-year movements in asset size and governance score do not by themselves indicate whether, or how strongly, female board representation contributed to the bank's financial trajectory.

Bank Asia

Table 11
Financial and Governance Indicators for Bank Asia (2016–2024)

Year	ROE %	ROA %	NIM %	CG Score %	Assets (mn BDT)
2024	9.04	0.55	5.14	73.52	532,901.82
2023	8.24	0.51	3.63	71.51	474,147.76
2022	10.88	0.67	3.27	71.18	468,400.94
2021	9.89	0.64	3.67	71.67	438,293.29
2020	7.81	0.53	2.73	74.28	408,960.69
2019	8.16	0.59	4.52	75.14	353,800.37
2018	10.21	0.76	4.34	73.59	307,291.40
2017	10.21	0.76	3.95	74.01	288,997.00
2016	8.13	0.65	4.2	73.14	253,196.00

Table 12
Board Gender Composition for Bank Asia (2016–2024)

Year	Board Total	Male	Female	Female %
2016	13	10	3	23.08
2017	12	9	3	25.0
2018	16	11	5	31.25
2019	13	9	4	30.77
2020	17	12	5	29.41
2021	18	13	5	27.78
2022	16	11	5	31.25
2023	15	11	4	26.67
2024	12	8	4	33.33

Across the reported period, Bank Asia shows ROE between 7.81% and 10.88% and ROA between 0.51% and 0.76%. Female board representation ranges from 23.08% to 33.33%. The bank-level pattern is therefore mixed: changes in profitability do not move in a uniform direction with female representation, and the governance score also changes from year to year. The figures

support comparison, but they do not by themselves prove that board gender composition caused the change in profitability.

Total assets for Bank Asia moved from approximately 253,196 million BDT in 2016 to approximately 532,902 million BDT in 2024, while the reported corporate governance score ranged from 71.18% to 75.14% over the same period. As with profitability, these year-to-year movements in asset size and governance score do not by themselves indicate whether, or how strongly, female board representation contributed to the bank's financial trajectory.

Prime Bank

Table 13

Financial and Governance Indicators for Prime Bank (2016–2024)

Year	ROE %	ROA %	NIM %	CG Score %	Assets (mn BDT)
2024	20.56	1.46	2.78	71.08	548,118.00
2023	14.33	1.06	3.0	64.28	473,091.00
2022	12.93	0.98	2.93	77.57	433,410.00
2021	10.61	0.84	3.18	78.44	389,878.00
2020	6.31	0.54	2.45	81.92	347,502.00
2019	5.93	0.76	3.81	86.9	323,788.00
2018	8.6	0.76	3.62	83.33	295,613.00
2017	4.24	0.38	2.56	83.95	281,467.00
2016	8.49	0.81	2.02	82.62	257,554.00

Table 14

Board Gender Composition for Prime Bank (2016–2024)

Year	Board Total	Male	Female	Female %
2016	19	16	3	15.79
2017	18	15	3	16.67
2018	18	15	3	16.67

2019	21	18	3	14.29
2020	19	17	2	10.53
2021	17	15	2	11.76
2022	20	18	2	10.0
2023	22	19	3	13.64
2024	19	16	4	21.05

Across the reported period, Prime Bank shows ROE between 4.24% and 20.56% and ROA between 0.38% and 1.46%. Female board representation ranges from 10.00% to 21.05%. The bank-level pattern is therefore mixed: changes in profitability do not move in a uniform direction with female representation, and the governance score also changes from year to year. The figures support comparison, but they do not by themselves prove that board gender composition caused the change in profitability.

Total assets for Prime Bank moved from approximately 257,554 million BDT in 2016 to approximately 548,118 million BDT in 2024, while the reported corporate governance score ranged from 64.28% to 86.90% over the same period. As with profitability, these year-to-year movements in asset size and governance score do not by themselves indicate whether, or how strongly, female board representation contributed to the bank's financial trajectory.

Southeast Bank

Table 15

Financial and Governance Indicators for Southeast Bank (2016–2024)

Year	ROE %	ROA %	NIM %	CG Score %	Assets (mn BDT)
2024	1.36	0.09	2.03	75.43	545,858.60
2023	6.8	0.42	2.58	76.96	510,864.24
2022	5.77	0.35	1.48	75.14	493,092.84

2021	5.98	0.37	1.62	72.09	495,841.96
2020	6.91	0.47	1.88	81.06	465,293.41
2019	8.56	0.62	2.42	87.87	422,313.00
2018	12.57	0.69	3.38	82.84	381,576.00
2017	4.46	0.37	3.15	87.5	339,288.00
2016	9.06	0.88	3.9	86.17	291,798.00

Table 16

Board Gender Composition for Southeast Bank (2016–2024)

Year	Board Total	Male	Female	Female %
2016	12	8	4	33.33
2017	12	8	4	33.33
2018	12	8	4	33.33
2019	12	9	3	25.0
2020	14	11	3	21.43
2021	13	10	3	23.08
2022	12	9	3	25.0
2023	11	8	3	27.27
2024	11	8	3	27.27

Across the reported period, Southeast Bank shows ROE between 1.36% and 12.57% and ROA between 0.09% and 0.88%. Female board representation ranges from 21.43% to 33.33%. The bank-level pattern is therefore mixed: changes in profitability do not move in a uniform direction with female representation, and the governance score also changes from year to year. The figures support comparison, but they do not by themselves prove that board gender composition caused the change in profitability.

Total assets for Southeast Bank moved from approximately 291,798 million BDT in 2016 to approximately 545,859 million BDT in 2024, while the reported corporate governance score ranged from 72.09% to 87.87% over the same period. As with profitability, these year-to-year

movements in asset size and governance score do not by themselves indicate whether, or how strongly, female board representation contributed to the bank's financial trajectory.

UCB

Table 17

Financial and Governance Indicators for UCB (2016–2024)

Year	ROE %	ROA %	NIM %	CG Score %	Assets (mn BDT)
2024	1.44	0.08	1.48	77.14	768,779.00
2023	6.41	0.41	0.66	81.54	685,012.00
2022	10.27	0.67	0.61	85.57	628,077.00
2021	8.76	0.6	1.07	83.81	565,314.00
2020	8.42	0.61	0.49	95.97	493,300.00
2019	8.84	0.65	1.21	87.71	446,760.00
2018	8.4	0.62	1.04	84.88	402,749.00
2017	9.31	0.7	1.74	88.23	363,315.00
2016	10.24	0.84	2.11	93.12	329,721.00

Table 18

Board Gender Composition for UCB (2016–2024)

Year	Board Total	Male	Female	Female %
2016	19	17	2	10.53
2017	20	19	1	5.0
2018	19	16	3	15.79
2019	21	18	3	14.29
2020	21	18	3	14.29
2021	21	18	3	14.29
2022	21	17	4	19.05
2023	21	17	4	19.05
2024	6	6	0	0.0

Across the reported period, UCB shows ROE between 1.44% and 10.27% and ROA between 0.08% and 0.84%. Female board representation ranges from 0.00% to 19.05%. The bank-level pattern is therefore mixed: changes in profitability do not move in a uniform direction with female representation, and the governance score also changes from year to year. The figures support comparison, but they do not by themselves prove that board gender composition caused the change in profitability.

Total assets for UCB moved from approximately 329,721 million BDT in 2016 to approximately 768,779 million BDT in 2024, while the reported corporate governance score ranged from 77.14% to 95.97% over the same period. As with profitability, these year-to-year movements in asset size and governance score do not by themselves indicate whether, or how strongly, female board representation contributed to the bank's financial trajectory.

Dhaka Bank

Table 19

Financial and Governance Indicators for Dhaka Bank (2016–2024)

Year	ROE %	ROA %	NIM %	CG Score %	Assets (mn BDT)
2024	5.71	0.31	3.9	97.72	431,459.00
2023	8.14	0.48	3.59	68.94	378,639.00
2022	8.09	0.49	2.54	74.34	346,556.00
2021	10.53	0.65	3.0	76.68	335,351.00
2020	11.28	0.7	2.97	78.81	295,337.00
2019	9.28	0.56	3.06	76.54	285,009.00
2018	8.18	0.54	3.79	74.23	273,976.00
2017	9.21	0.69	3.9	80.23	229,453.00
2016	10.15	0.77	4.58	84.0	202,192.00

Table 20

Board Gender Composition for Dhaka Bank (2016–2024)

Year	Board Total	Male	Female	Female %
2016	17	16	1	5.88
2017	17	16	1	5.88
2018	18	16	2	11.11
2019	18	16	2	11.11
2020	18	15	3	16.67
2021	18	15	3	16.67
2022	18	15	3	16.67
2023	18	15	3	16.67
2024	18	15	4	22.22

Across the reported period, Dhaka Bank shows ROE between 5.71% and 11.28% and ROA between 0.31% and 0.77%. Female board representation ranges from 5.88% to 22.22%. The bank-level pattern is therefore mixed: changes in profitability do not move in a uniform direction with female representation, and the governance score also changes from year to year. The figures support comparison, but they do not by themselves prove that board gender composition caused the change in profitability.

Total assets for Dhaka Bank moved from approximately 202,192 million BDT in 2016 to approximately 431,459 million BDT in 2024, while the reported corporate governance score ranged from 68.94% to 97.72% over the same period. As with profitability, these year-to-year movements in asset size and governance score do not by themselves indicate whether, or how strongly, female board representation contributed to the bank's financial trajectory.

Eastern Bank PLC

Table 21

Financial and Governance Indicators for Eastern Bank PLC (2016–2024)

Year	ROE %	ROA %	NIM %	CG Score %	Assets (mn BDT)
2024	18.63	1.34	3.33	88.88	616,050.00
2023	16.85	1.27	3.23	95.51	506,271.00
2022	15.46	1.21	3.28	92.85	455,989.00
2021	15.51	1.28	3.85	78.48	388,815.00
2020	15.04	1.22	3.15	73.78	336,936.00
2019	16.52	1.3	4.67	69.11	335,163.00
2018	13.83	1.15	4.9	68.9	282,451.00
2017	11.41	1.04	4.41	65.77	253,068.00
2016	12.94	1.33	5.11	61.8	211,185.00

Table 22

Board Gender Composition for Eastern Bank PLC (2016–2024)

Year	Board Total	Male	Female	Female %
2016	11	10	1	9.09
2017	11	10	1	9.09
2018	11	10	1	9.09
2019	11	10	1	9.09
2020	11	8	3	27.27
2021	13	10	3	23.08
2022	14	11	3	21.43
2023	14	12	2	14.29
2024	13	9	4	30.77

Across the reported period, Eastern Bank PLC shows ROE between 11.41% and 18.63% and ROA between 1.04% and 1.34%. Female board representation ranges from 9.09% to 30.77%. The bank-level pattern is therefore mixed: changes in profitability do not move in a uniform direction with female representation, and the governance score also changes from year

to year. The figures support comparison, but they do not by themselves prove that board gender composition caused the change in profitability.

Total assets for Eastern Bank PLC moved from approximately 211,185 million BDT in 2016 to approximately 616,050 million BDT in 2024, while the reported corporate governance score ranged from 61.80% to 95.51% over the same period. As with profitability, these year-to-year movements in asset size and governance score do not by themselves indicate whether, or how strongly, female board representation contributed to the bank's financial trajectory.

6. Cross-Bank Findings and Discussion

The bank-level evidence shows that profitability and female board representation move differently across institutions. This is important because it prevents a simple interpretation of the relationship. For example, BRAC Bank, City Bank, and Bank Asia show periods in which female representation is relatively high and profitability is also comparatively strong. However, Southeast Bank and UCB provide counter-examples in which female representation or governance scores can remain substantial while profitability declines. These differences indicate that bank-specific and macro-level factors matter alongside gender diversity.

Table 23

Summary of Female Participation and Governance Scores Across Banks

Bank	Highest female representation	Lowest female representation	Highest CG score	Lowest CG score
Dutch-Bangla Bank	25.00% (2023–2024)	0% (2016–2018)	81.44% (2016)	66.87% (2020)
BRAC Bank	42.86% (2019)	20.00% (2021–2022)	97.91% (2017)	72.39% (2024)
City Bank	36.36% (2022)	17.65% (2016–2017)	99.96% (2023)	74.06% (2019)
Bank Asia	33.33% (2024)	23.07% (2016)	75.14% (2019)	71.18% (2022)
Prime Bank	21.05% (2024)	10.00% (2022)	86.90% (2019)	64.28% (2023)

Southeast Bank	33.33% (2016–2018)	21.43% (2020)	87.87% (2019)	72.09% (2021)
UCB	19.05% (2022–2023)	0% (2024)	95.97% (2020)	77.14% (2024)
Dhaka Bank	22.22% (2024)	5.88% (2016–2017)	97.72% (2024)	68.94% (2023)
Eastern Bank PLC	30.77% (2024)	9.09% (2016–2019)	95.51% (2023)	61.80% (2016)

6.1 Profitability Patterns

- ROE shows wide variation across banks. City Bank reaches the highest reported ROE of 26.1% in 2024, while Southeast Bank records the lowest reported ROE of 1.36% in 2024.
- ROA is generally more stable than ROE. The highest reported ROA is 2.02% for BRAC Bank in 2017, while the lowest is 0.08% for UCB in 2024.
- NIM varies widely in the submitted data. City Bank reports unusually high values in 2016–2019, so these observations should be checked against the original annual reports before being used for strong cross-bank comparisons.
- Profitability is not explained by one governance variable. The contrasting paths of the nine banks show that operational efficiency, asset quality, market conditions, funding costs, and other bank-specific factors can also matter.

6.2 Corporate Governance Patterns

Governance scores differ substantially across banks and across years. Some banks show high reported compliance scores, yet profitability does not always rise at the same time. This weakens the idea of a simple one-directional relationship between governance score and profitability. The result is more consistent with a broader view in which governance provides an institutional foundation, while financial performance is shaped by several interacting factors.

6.3 Female Participation Patterns

Female representation increased or remained meaningful in several of the selected banks. BRAC Bank, City Bank, Bank Asia, Dhaka Bank, and Eastern Bank show years with relatively higher female representation than their own earlier observations. At the same time, UCB's reported 2024 board composition contains no female members, and Dutch-Bangla Bank reports no female board members from 2016 to 2018. These differences demonstrate that women's participation in Bangladeshi banking governance is uneven across institutions.

6.4 Does the Evidence Show a Critical Mass?

The draft paper originally treated three women as a likely critical mass. The revised interpretation is more cautious. Several banks record three or more women in some years, but the descriptive patterns do not show a consistent profitability improvement whenever that threshold is reached. The regression's squared female term is positive with $p = 0.0672$, which is suggestive of non-linearity but does not establish a universal three-person threshold. A stronger test would require a clearly defined non-linear panel model, a documented sample size, bank fixed effects, time fixed effects, and robustness checks.

7. Results and Discussion

The first broad finding is that profitability differs substantially across banks and years. This is expected in a banking sample because banks differ in scale, business mix, operating efficiency, credit risk, interest-rate exposure, and market position. The data therefore support a multi-factor view of profitability rather than a single-variable explanation.

Second, the study does not find a statistically significant linear effect of female participation in the reported regression at the 5% level. The negative coefficient on the female-cumulative term is not statistically significant. This means the coefficient should not be

interpreted as evidence that women's participation reduces profitability. A non-significant coefficient means that the submitted model does not provide sufficiently strong evidence of a linear effect in that direction.

Third, the positive coefficient on the squared female term provides some evidence consistent with a non-linear relationship. However, the p-value of 0.0672 places the result above the conventional 5% threshold. It is therefore more appropriate to describe this as suggestive rather than conclusive. The finding is useful for future research because it indicates that the effect of representation may depend on the level of participation, but the present study cannot establish exactly where a threshold occurs.

Fourth, the high correlations among ROE, ROA, NIM, governance score, and bank size require caution. A high correlation can create multicollinearity when several variables enter the same regression model. The original paper does not report a formal variance-inflation test or other multicollinearity diagnostics, so this remains an unresolved methodological issue.

Finally, the cross-bank patterns suggest that governance quality and gender diversity should be treated as part of a larger governance system rather than as isolated drivers of profit. Women's participation may be valuable for representation, monitoring, risk discussion, stakeholder understanding, and board dynamics, but the available data do not support a claim that gender composition alone determines bank profitability.

7.1 Visual Analysis of the Reported Data

The original submission contained several graphs and gender-distribution figures. These visuals have been retained in the revised version so that the reader can see the reported trends directly. No values were changed in the extracted figures. They should be read together with the tables and the data-quality notes in Section 10.

Dutch-Bangla Bank

Figure 1

ROE, ROA and NIM Trends for Dutch-Bangla Bank (2016–2024)

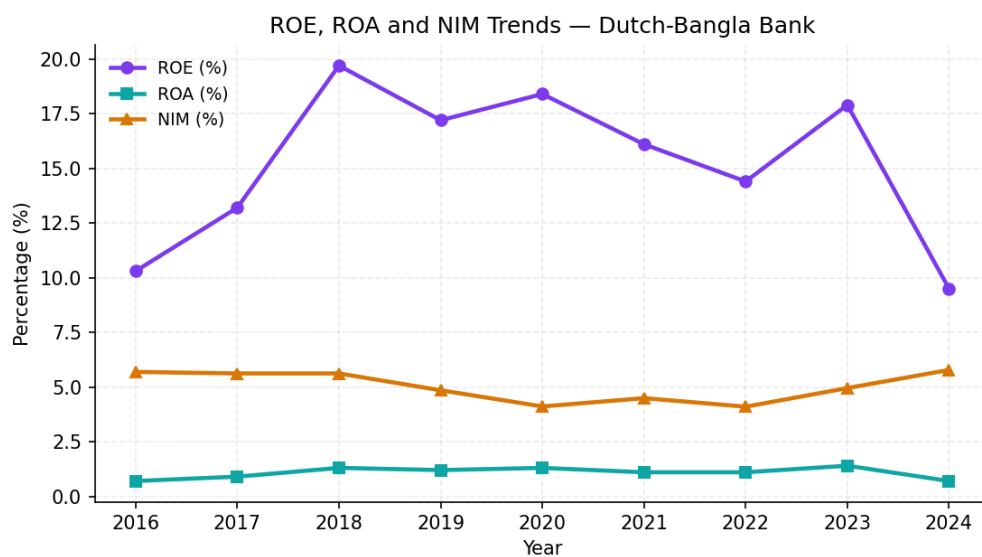


Figure 2

Gender Distribution of the Dutch-Bangla Bank Board (2016–2024)

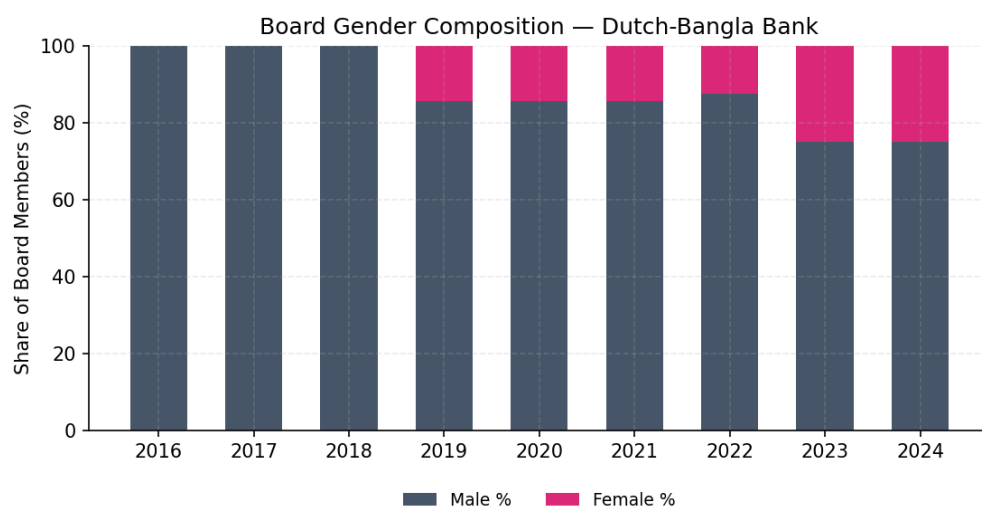
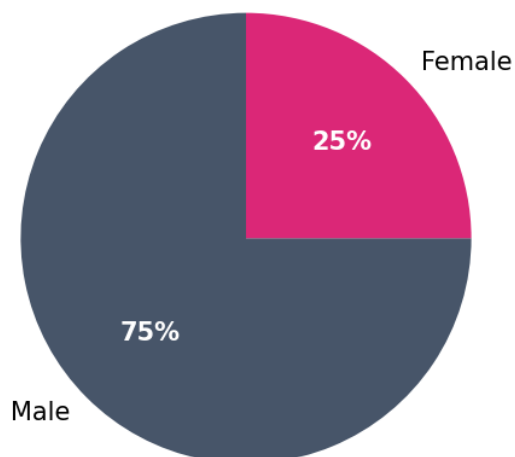


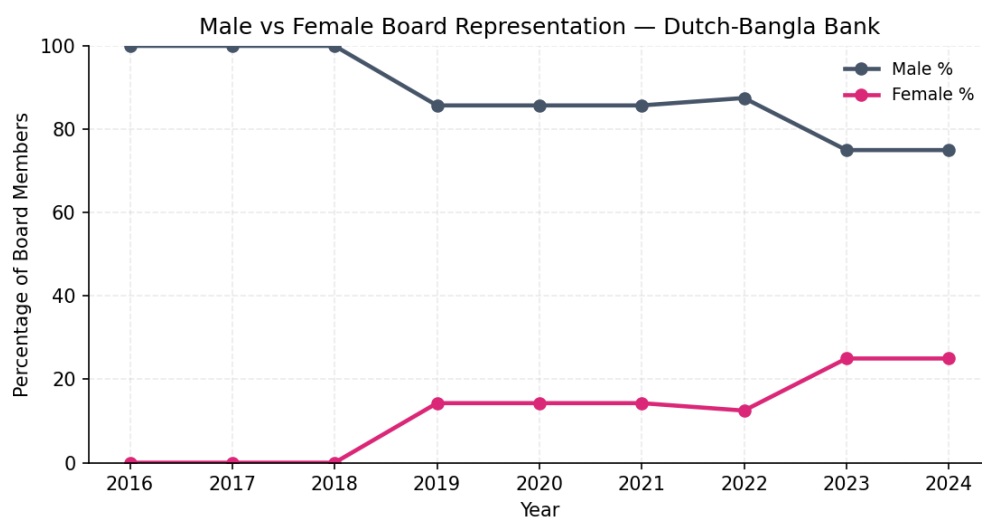
Figure 3

Gender Split in 2024 — Dutch-Bangla Bank

Gender Split in 2024 — Dutch-Bangla Bank

**Figure 4**

Male and Female Board Representation at Dutch-Bangla Bank (2016–2024)



BRAC Bank

Figure 5

ROE, ROA and NIM Trends for BRAC Bank (2016–2024)

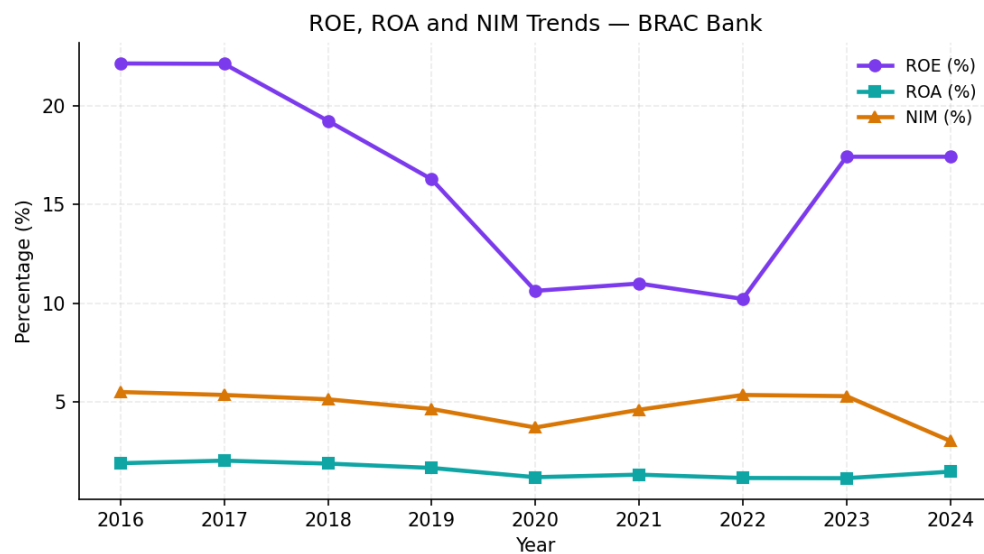


Figure 6
Gender Distribution of the BRAC Bank Board (2016–2024)

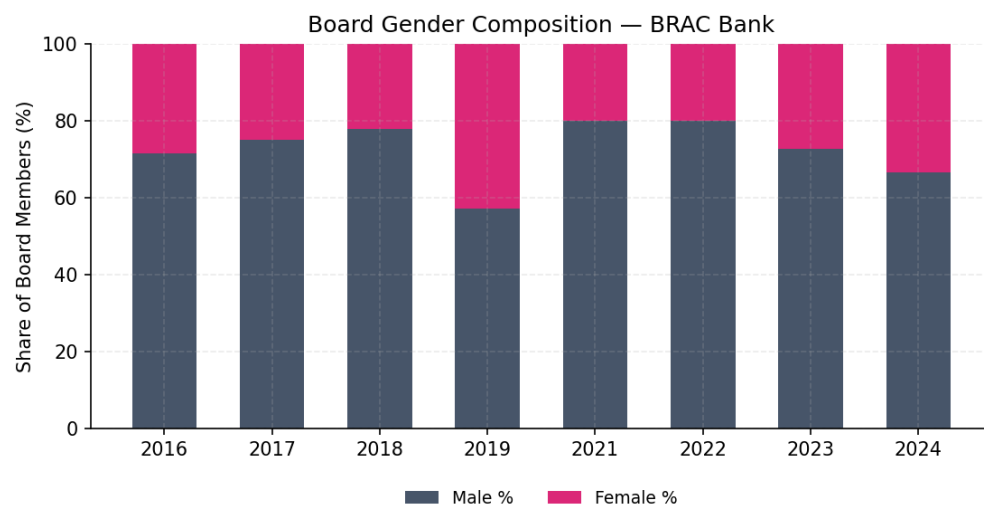
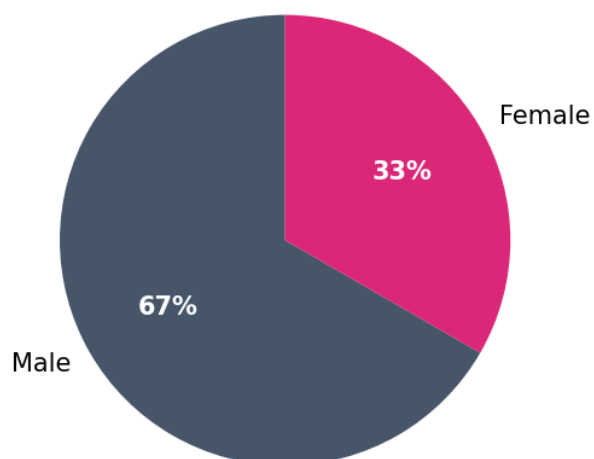


Figure 7
Gender Split in 2024 — BRAC Bank

Gender Split in 2024 — BRAC Bank



City Bank

Figure 8

ROE, ROA and NIM Trends for City Bank (2016–2024)

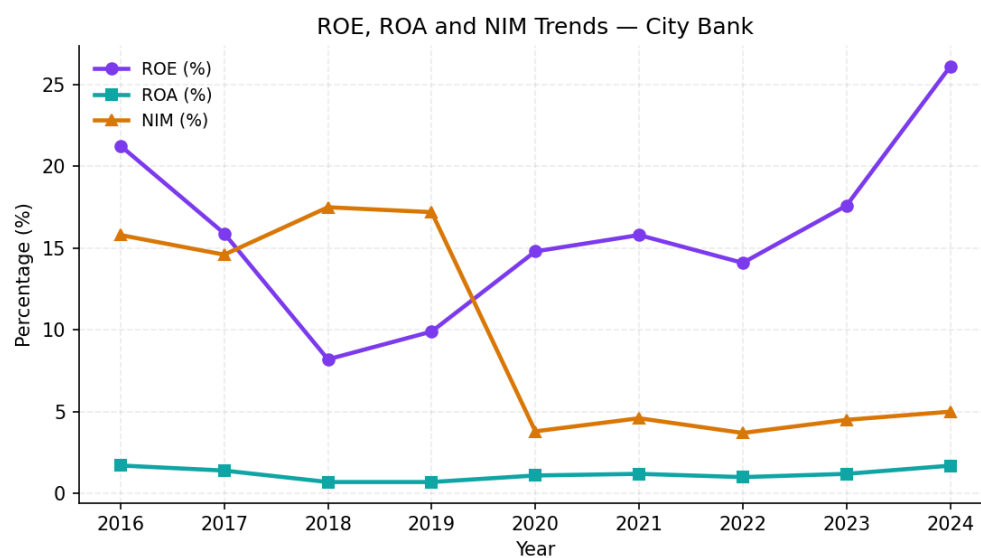


Figure 9

Gender Distribution of the City Bank Board (2016–2024)

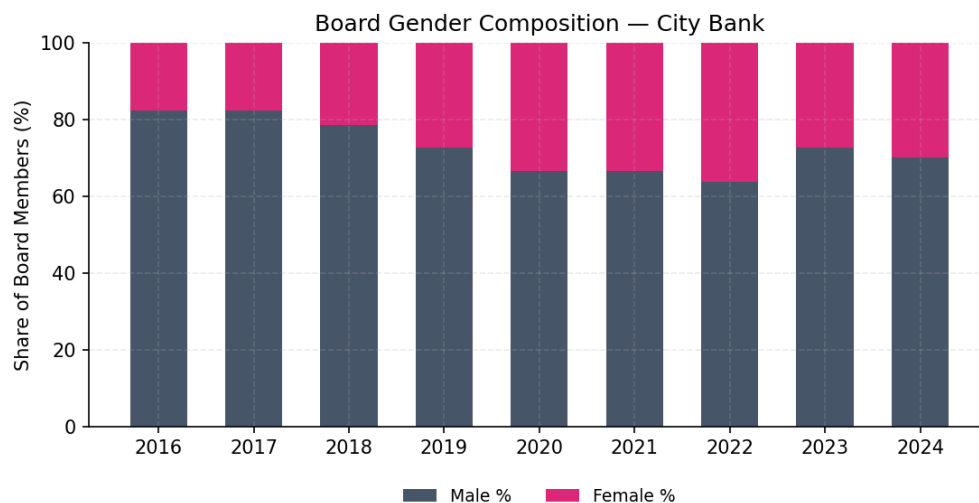
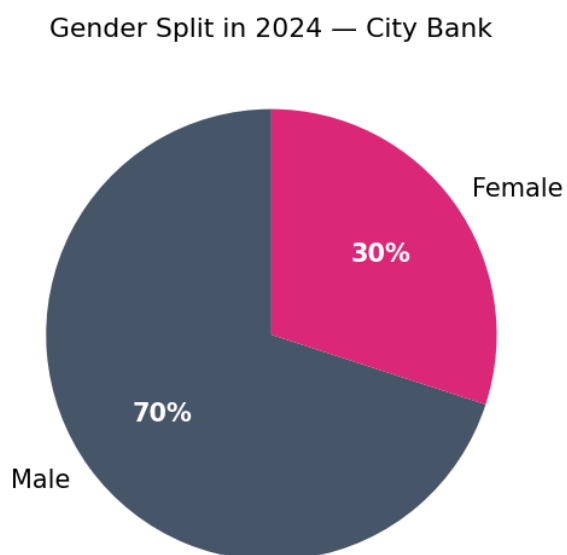


Figure 10
Gender Split in 2024 — City Bank



Bank Asia

Figure 11
ROE, ROA and NIM Trends for Bank Asia (2016–2024)

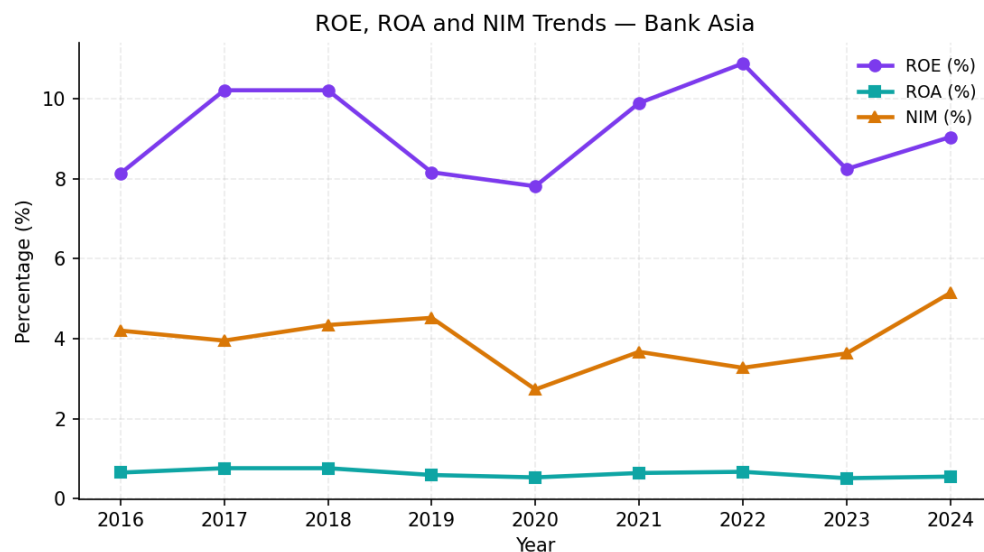


Figure 12
Gender Distribution of the Bank Asia Board (2016–2024)

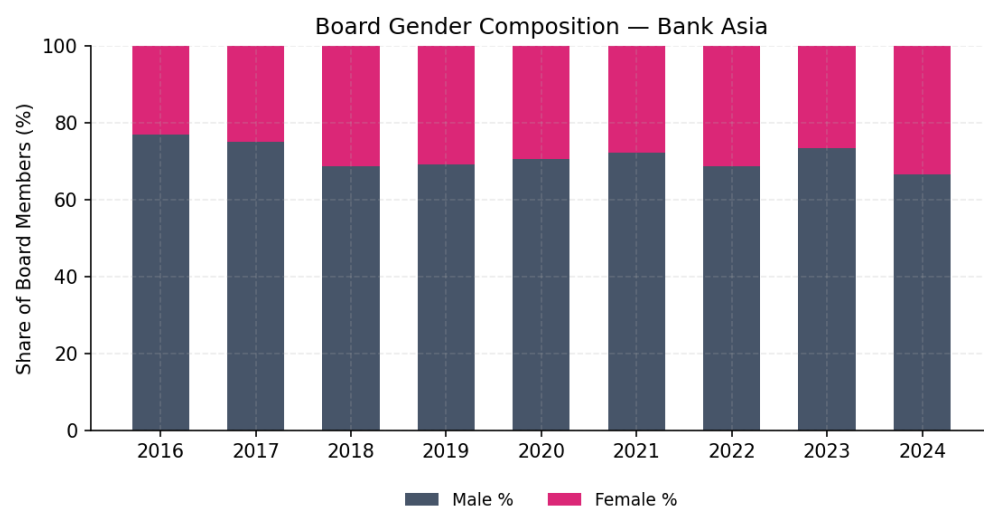


Figure 13
Gender Split in 2024 — Bank Asia

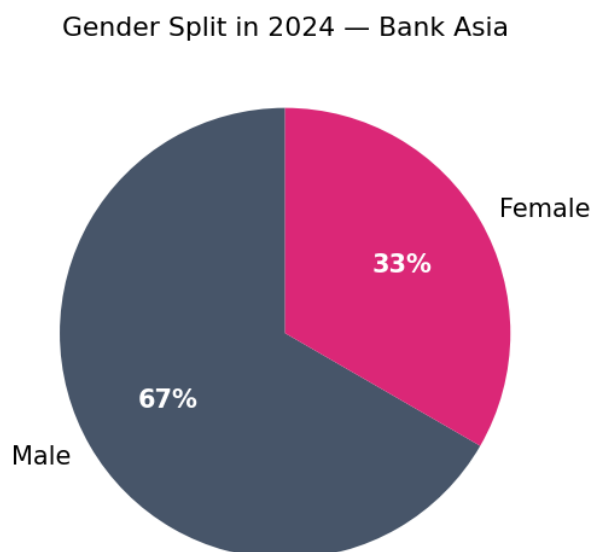
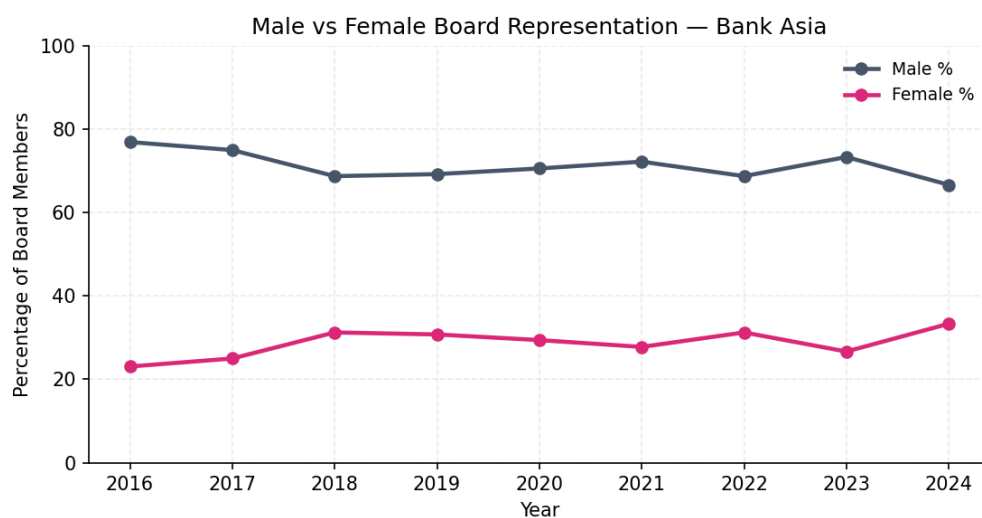


Figure 14
Male and Female Board Representation at Bank Asia (2016–2024)



Prime Bank

Figure 15
ROE, ROA and NIM Trends for Prime Bank (2016–2024)

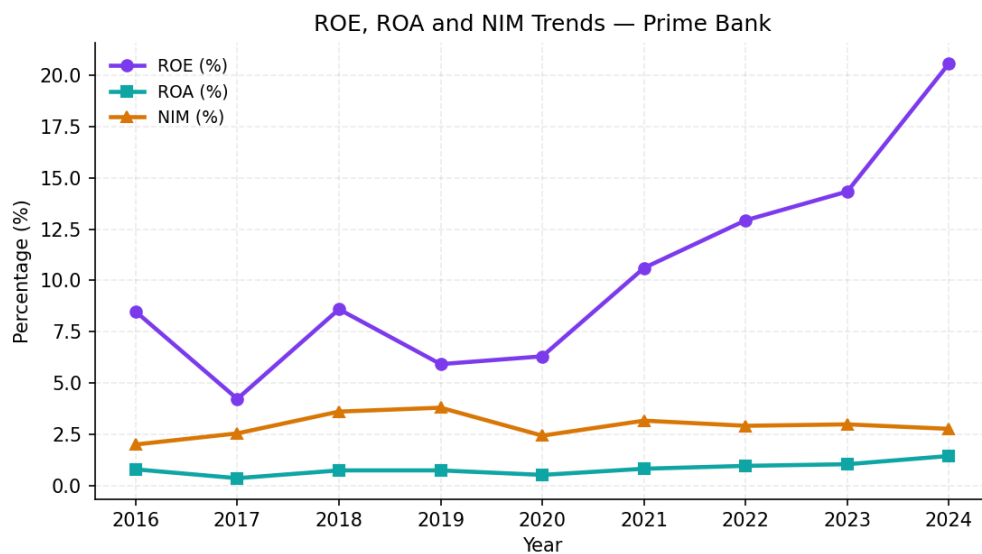


Figure 16
Gender Distribution of the Prime Bank Board (2016–2024)

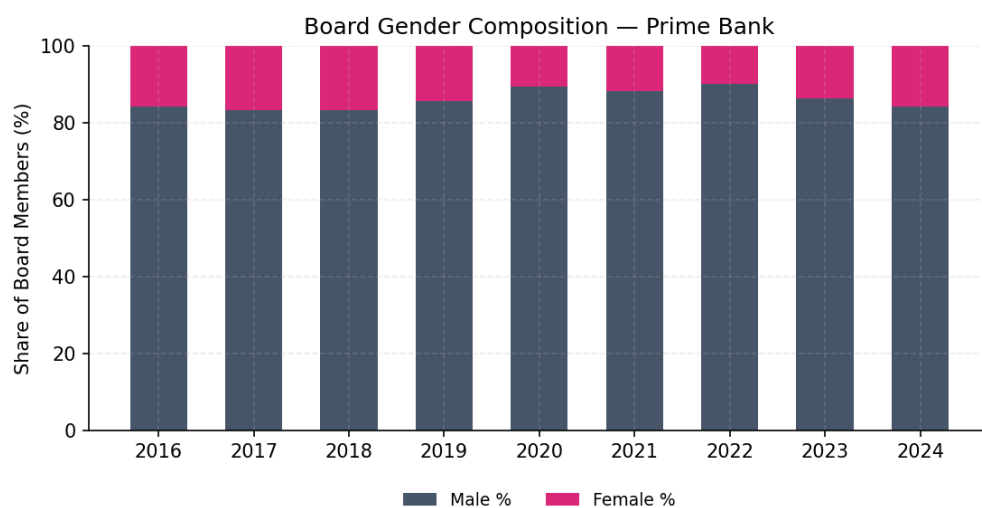
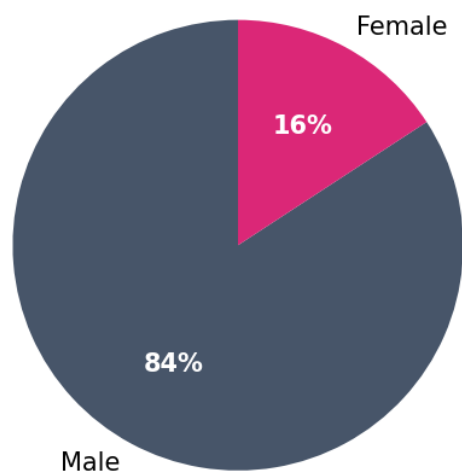
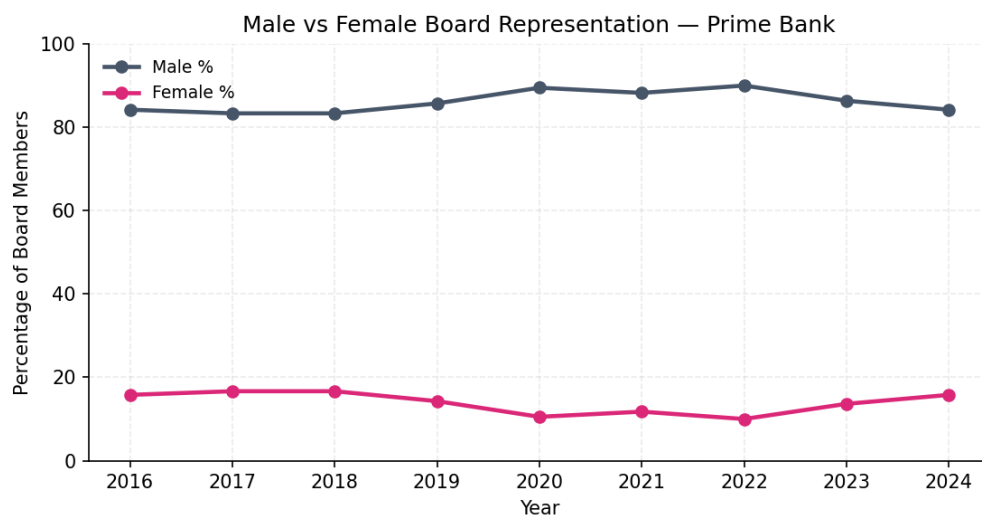


Figure 17
Gender Split in 2024 — Prime Bank

Gender Split in 2024 — Prime Bank

**Figure 18**

Male and Female Board Representation at Prime Bank (2016–2024)



Southeast Bank

Figure 19

ROE, ROA and NIM Trends for Southeast Bank (2016–2024)

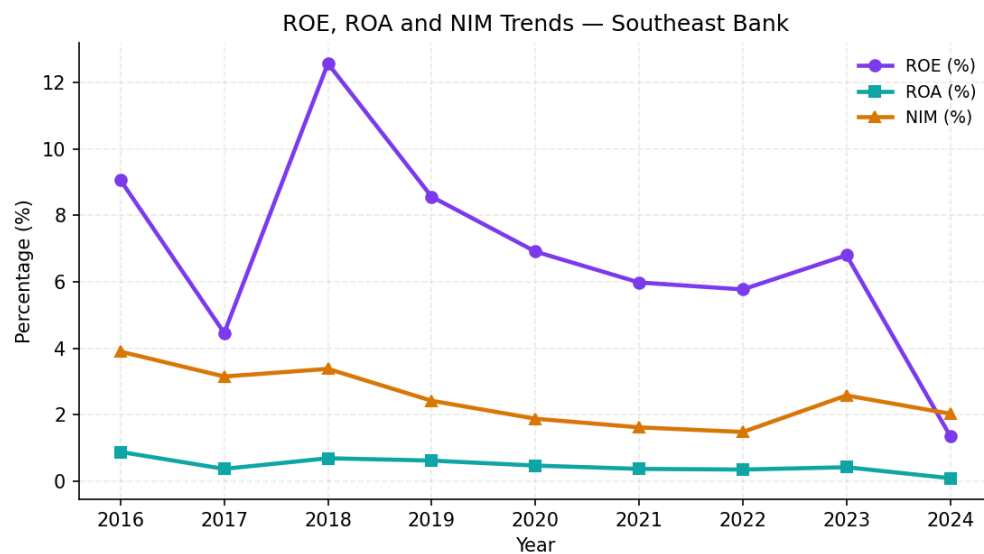


Figure 20
Gender Distribution of the Southeast Bank Board (2016–2024)

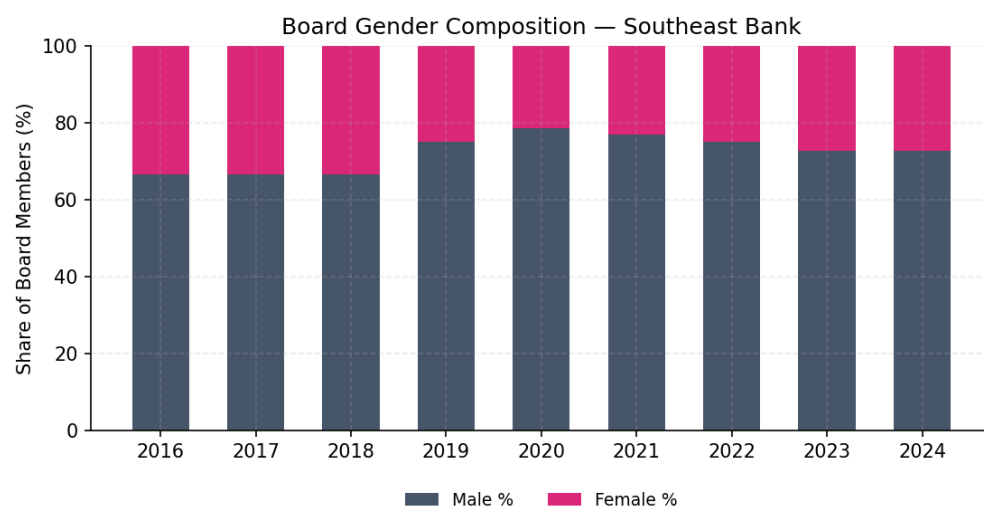
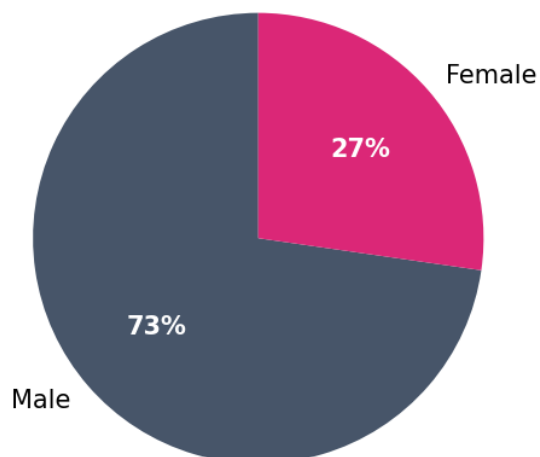
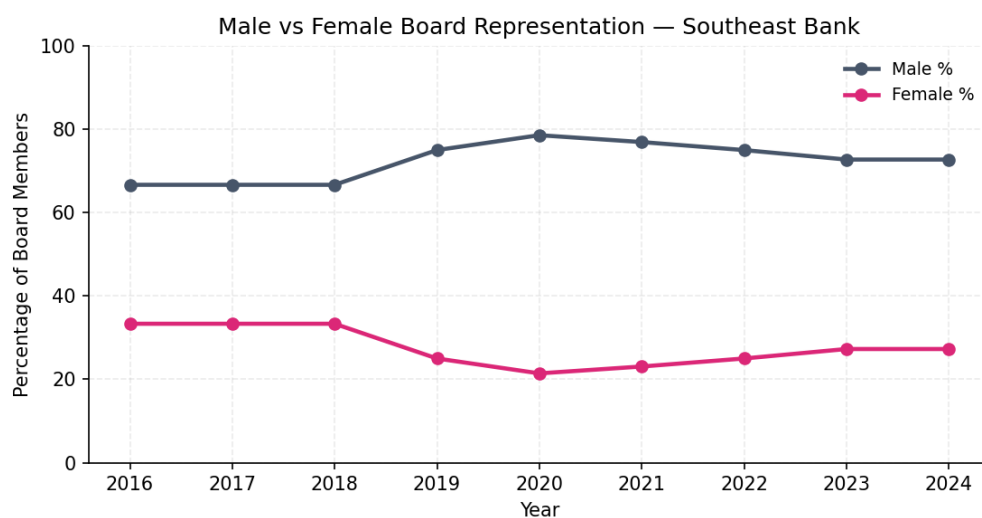


Figure 21
Gender Split in 2024 — Southeast Bank

Gender Split in 2024 — Southeast Bank

**Figure 22**

Male and Female Board Representation at Southeast Bank (2016–2024)



United Commercial Bank (UCB)

Figure 23

ROE, ROA and NIM Trends for United Commercial Bank (UCB) (2016–2024)

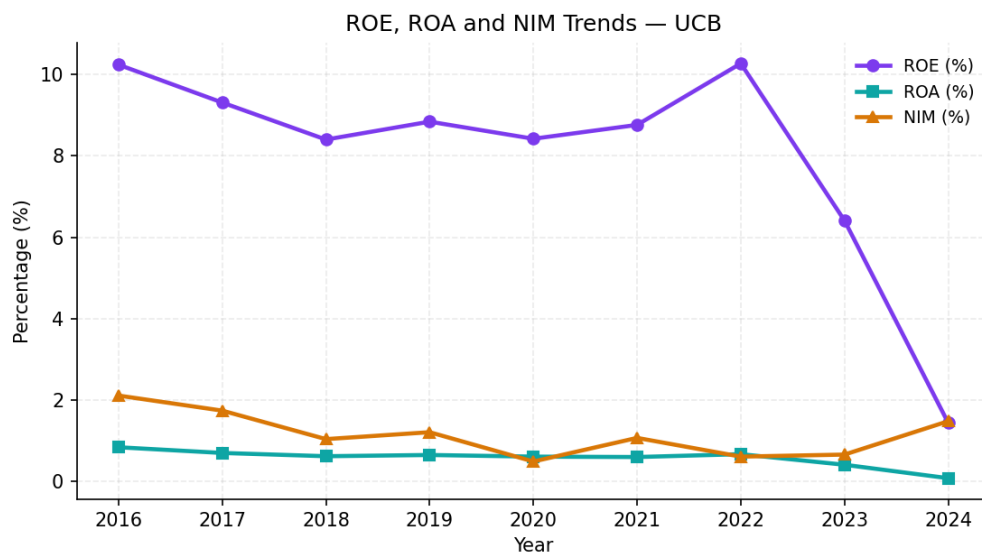


Figure 24

Gender Distribution of the United Commercial Bank (UCB) Board (2016–2024)

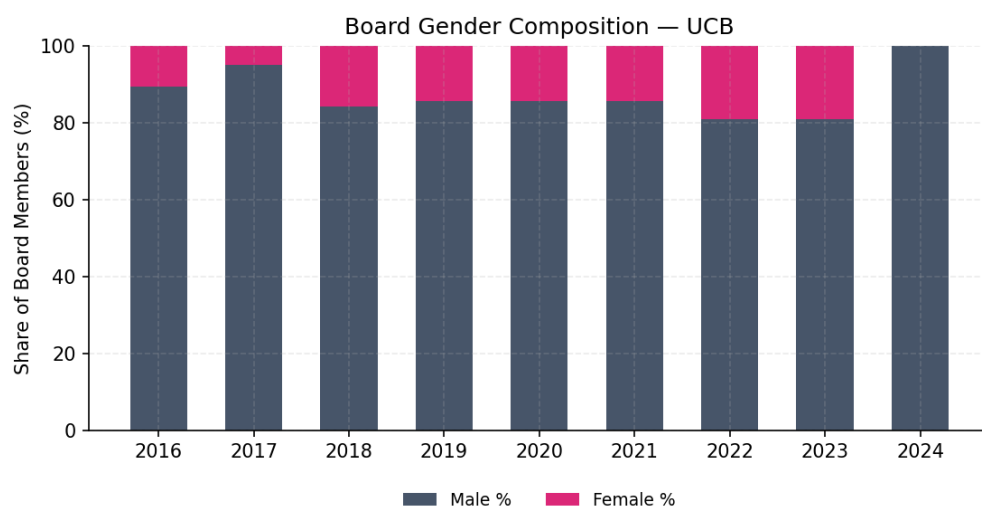


Figure 25

Gender Split in 2024 — United Commercial Bank (UCB)

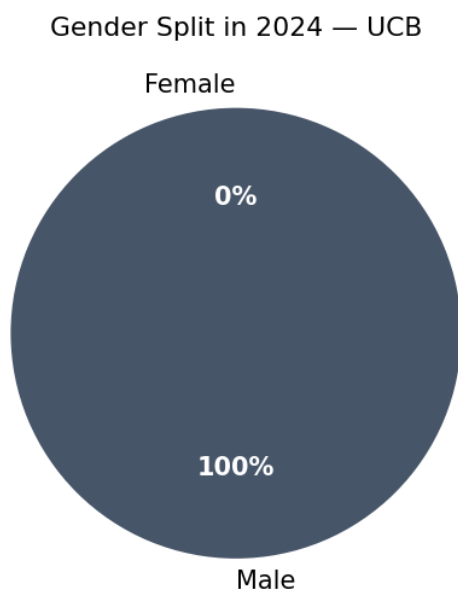
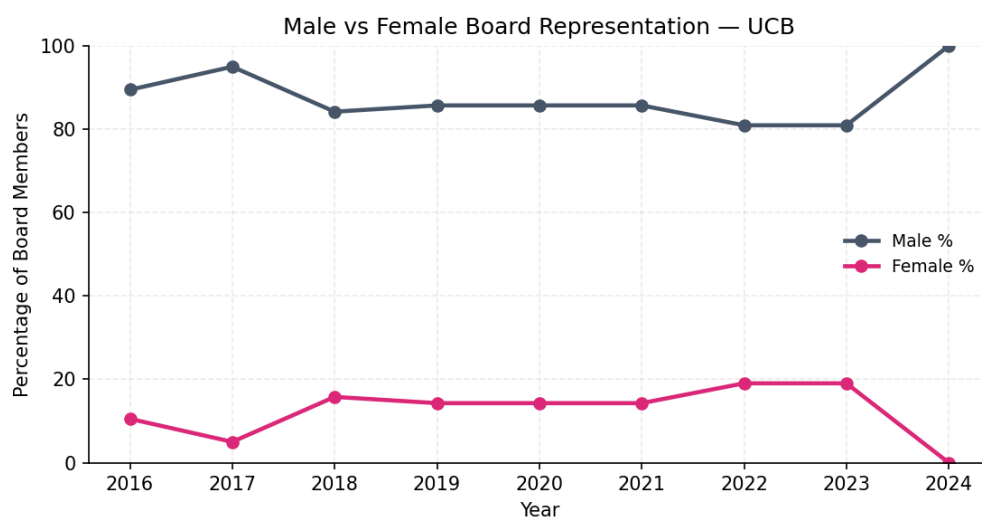


Figure 26

Male and Female Board Representation at United Commercial Bank (UCB) (2016–2024)



Dhaka Bank

Figure 27

ROE, ROA and NIM Trends for Dhaka Bank (2016–2024)

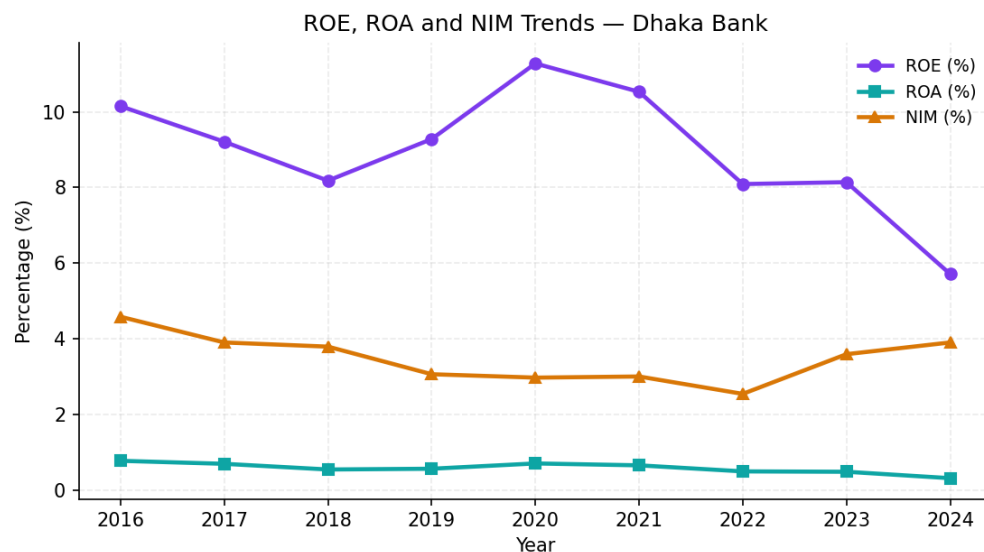


Figure 28
Gender Distribution of the Dhaka Bank Board (2016–2024)

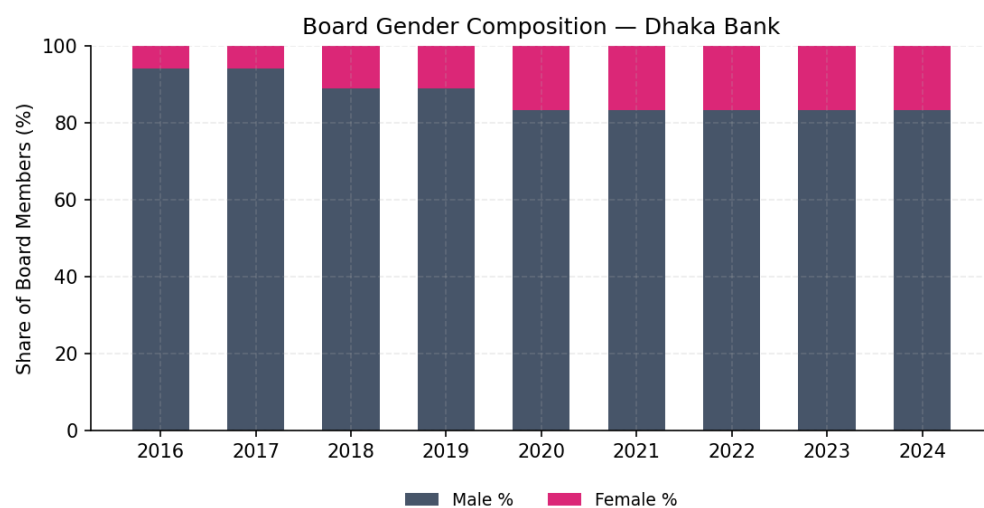
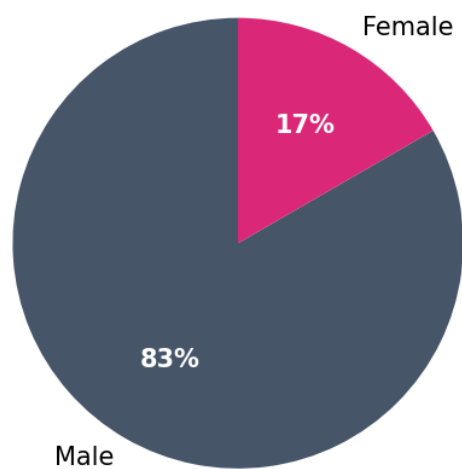
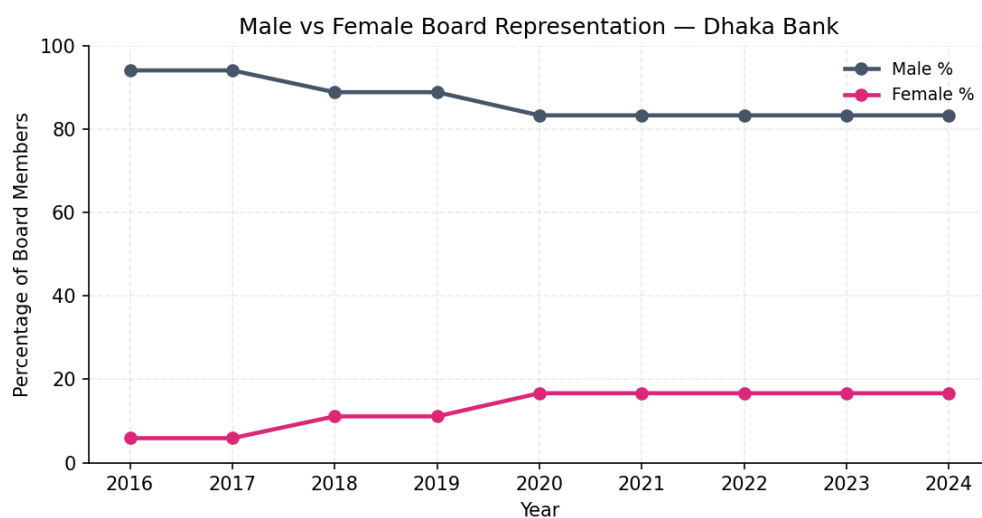


Figure 29
Gender Split in 2024 — Dhaka Bank

Gender Split in 2024 — Dhaka Bank

**Figure 30**

Male and Female Board Representation at Dhaka Bank (2016–2024)



Eastern Bank PLC

Figure 31

ROE, ROA and NIM Trends for Eastern Bank PLC (2016–2024)

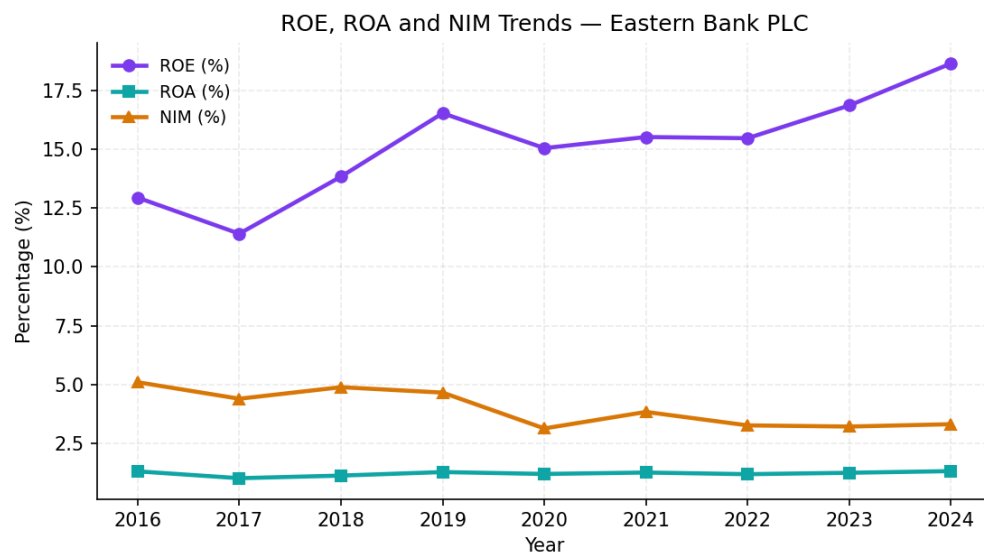


Figure 32
Gender Distribution of the Eastern Bank PLC Board (2016–2024)

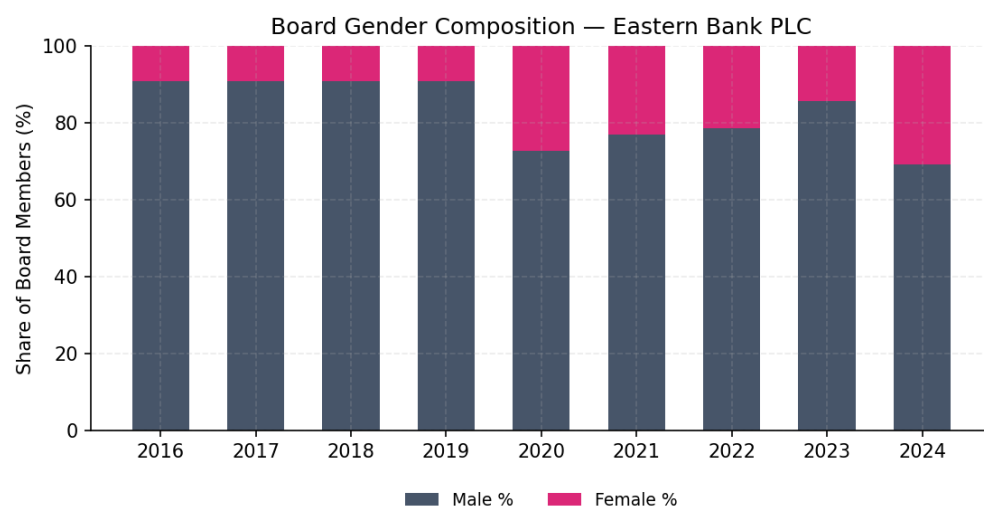
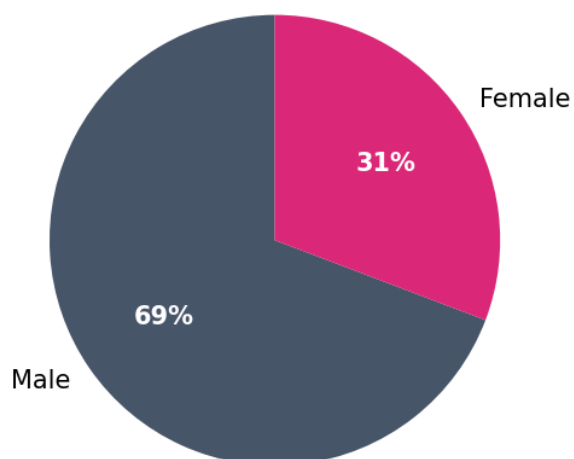
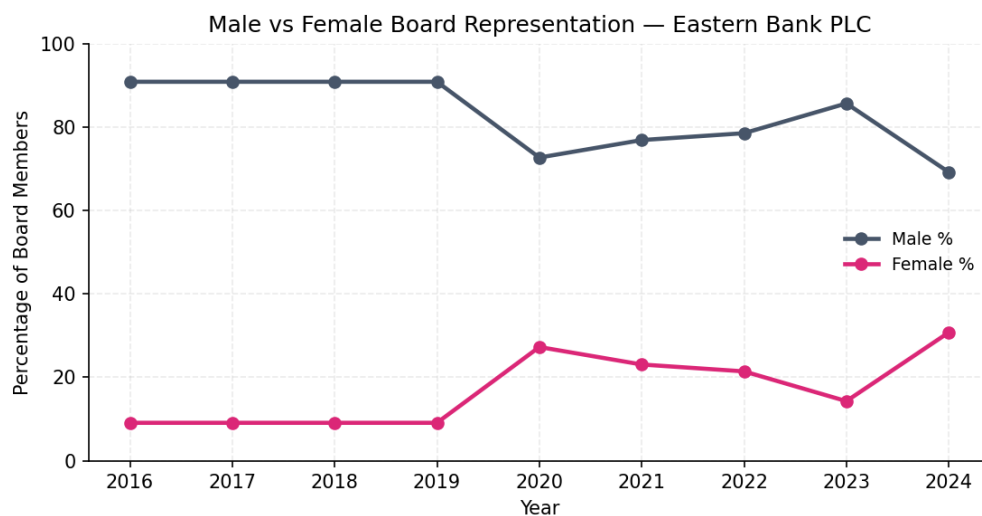


Figure 33
Gender Split in 2024 — Eastern Bank PLC

Gender Split in 2024 — Eastern Bank PLC

**Figure 34**

Male and Female Board Representation at Eastern Bank PLC (2016–2024)



Cross-Bank Comparison

Figure 35

Average ROE (2016–2024) — trend across selected banks

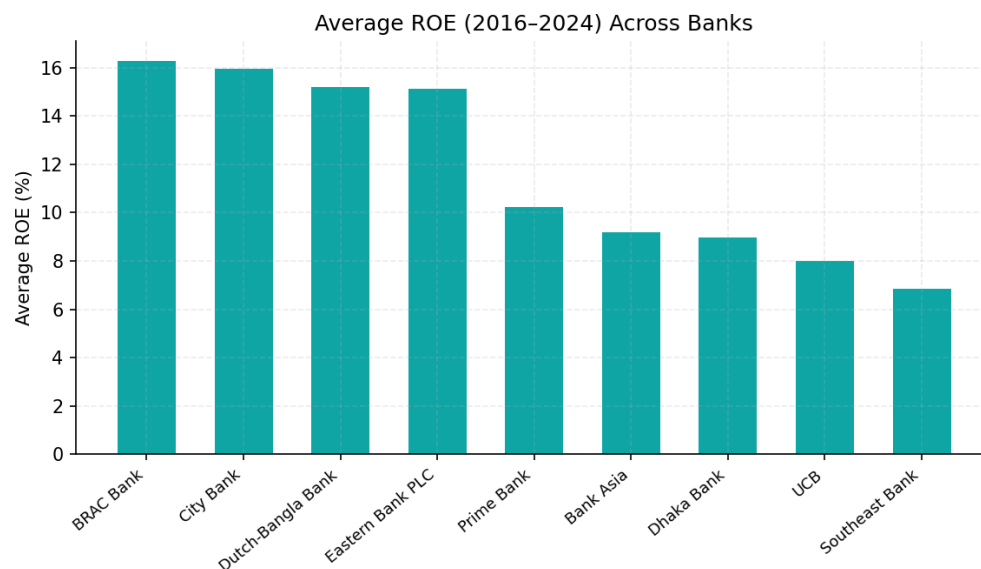


Figure 36
Average ROA (2016–2024) — trend across selected banks

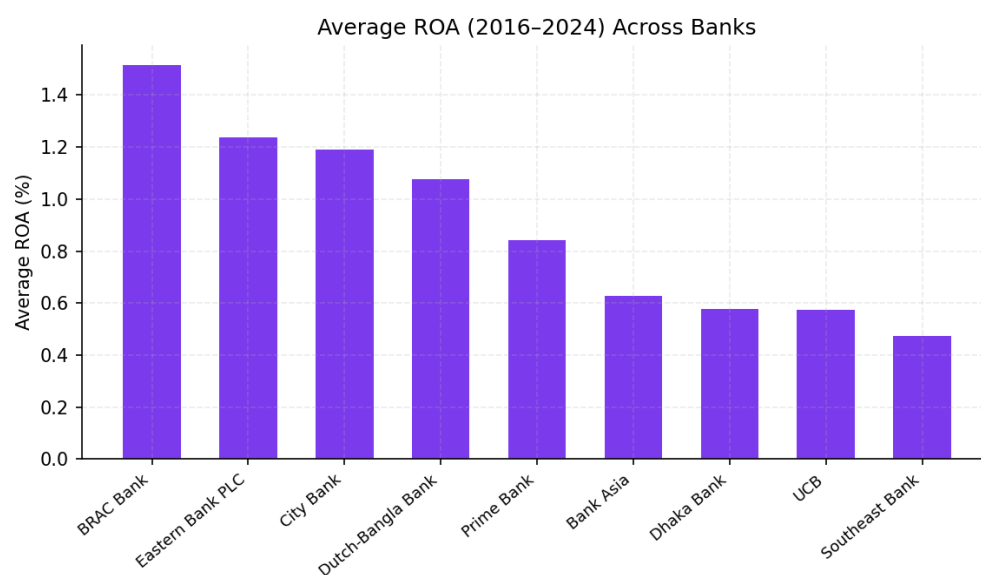
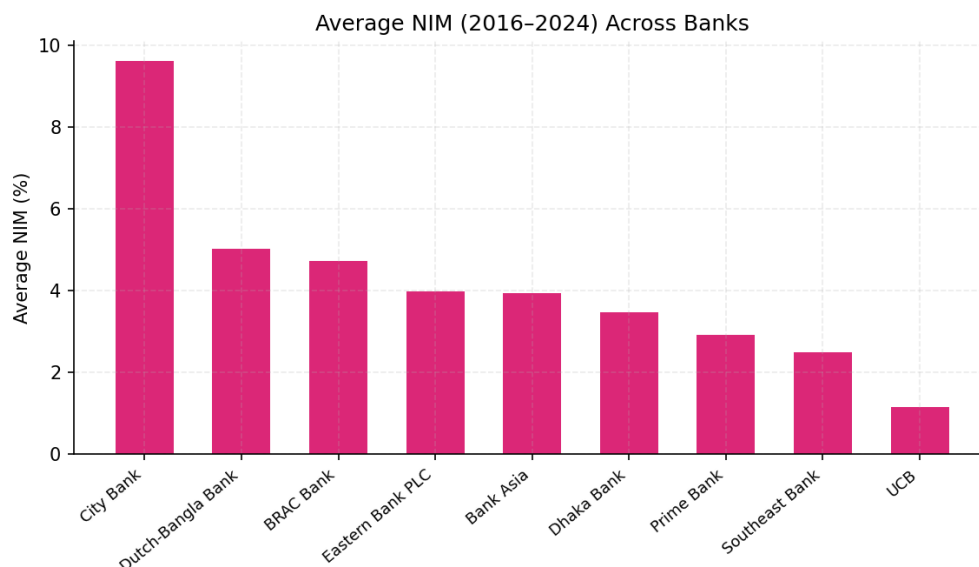


Figure 37
Average NIM (2016–2024) — trend across selected banks



Note: The graphs reproduce the figures present in the submitted PDF. Where the underlying source data were flagged as inconsistent in Section 10, the figure is retained for transparency rather than silently altered.

8. Practical Implications

For bank boards, the main implication is that increasing female participation should be connected to meaningful involvement in governance rather than treated only as a numerical target. Directors should have appropriate committee responsibilities, access to information, and a genuine role in monitoring and strategic discussion.

For regulators, board-composition monitoring can be combined with broader governance indicators such as independence, committee effectiveness, disclosure quality, risk oversight, and compliance. The evidence in this paper suggests that gender diversity is relevant, but it should be evaluated together with these broader governance mechanisms.

For banks, stronger disclosure of board composition and senior-management diversity would improve future research. Consistent annual reporting would make it easier to compare

banks over time and test whether changes in representation are associated with changes in profitability, risk, or stability.

9. Limitations of the Study

- The study covers only nine private commercial banks in Bangladesh, so the results should not be generalized to all banks or other countries.
- The submitted dataset contains inconsistencies and at least one missing bank-year board observation. These issues reduce reproducibility unless the original annual reports are rechecked.
- The reported regression table does not provide enough information about sample size, model fit, fixed effects implementation, or robustness tests to reproduce the full statistical analysis.
- The paper does not provide sufficient evidence to establish causality. Correlation and regression coefficients from the supplied specification should be interpreted as associations.
- Some variables are not consistently defined across sections, particularly NIM and management diversity. The final submitted version should use one operational definition throughout.

10. Data Quality and Consistency Notes

The following points should be checked against the original annual reports before the paper is submitted as a final research paper:

- BRAC Bank's board table omits 2020 even though the study describes a 2016–2024 nine-year period.

- City Bank's 2016 asset figure is reported as 25,940.23 million BDT in the financial table, while the accompanying narrative describes approximately 275,940 million BDT. The source figure should be verified before calculating growth rates.
- The descriptive-statistics table has count = 9, which does not represent the full nine-bank by nine-year panel. The authors should state exactly what observations were used to generate that table.
- The original conclusion states that governance variables have a positive and meaningful effect, but the reported p-values for corporate governance and bank size are not statistically significant at the 5% level. The conclusion has therefore been rewritten to match the reported evidence.
- The original discussion states that critical mass is usually three women. The revised paper treats this only as a theoretical proposition because the submitted regression does not establish a universal threshold.
- City Bank's unusually high NIM values in 2016–2019 should be verified against the bank's annual reports because they materially affect cross-bank comparisons.

11. Conclusion

This study examined corporate governance, female participation, and profitability across nine selected private commercial banks in Bangladesh during 2016–2024. The descriptive evidence shows clear differences in profitability, governance scores, asset size, and female board representation across banks. Some banks combined relatively strong profitability with relatively higher female participation, but other banks showed that profitability can decline even when female representation or governance scores remain comparatively strong.

The reported regression results do not support a simple conclusion that more women on bank boards directly increase profitability. The linear female-participation coefficient is statistically insignificant, while the squared term is positive and only marginally significant at the 10% level. This suggests that a non-linear relationship may be worth investigating, but it does not prove a specific critical mass or a causal profitability effect.

The main contribution of the study is therefore not a claim that women's participation automatically produces higher profit. Instead, the evidence highlights the importance of examining board diversity as part of a wider corporate-governance system. Female representation can potentially broaden perspectives, strengthen oversight, and improve stakeholder representation, but the actual financial outcome depends on the broader institutional and operational context of each bank.

Future studies should use a fully verified bank-year panel, clearly defined variables, bank and year fixed effects, larger samples where possible, and robustness tests. This would provide a stronger basis for determining whether female participation affects profitability, risk-taking, or stability in Bangladesh's banking sector.

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